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    (polynomial_features, polynomial_features),
    ("linear_regression", linear_regression),
})

# Train model
pipeline.fit(X[:, np.newaxis], y)

# Cross-validation
scores = cross_val_score(
    pipeline,
    X[:, np.newaxis],
    y,
    scoring="neg_mean_squared_error",
    cv=10
)

# Test data
X_test = np.linspace(0, 1, 100)

# Plot
plt.plot(X_test, pipeline.predict(X_test[:, np.newaxis]), label="Model")
plt.plot(X_test, true_fun(X_test), label="True Function")
plt.scatter(X, y, edgecolor="y", s=20, marker="^", label="Samples")

plt.xlabel("x")
plt.ylabel("y")
plt.xlim((0, 1))
plt.ylim((-2, 2))
plt.legend(loc="best")

plt.title(
    "Degree {} \n MSE = {:.2e} (+/- {:.2e})".format(
        degrees[i],
        -scores.mean(),
        scores.std()
    )
)

plt.tight_layout()
plt.show()

```

